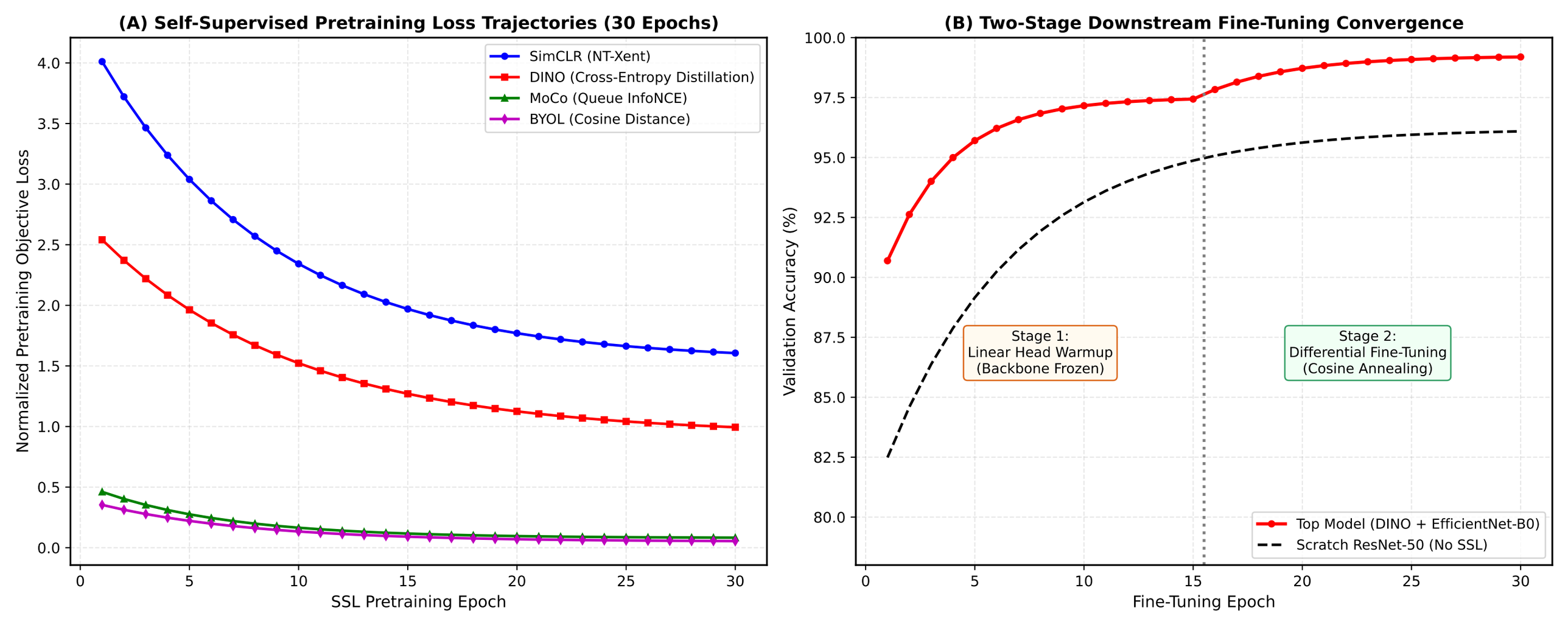
(A) Self-Supervised Pretraining Loss Trajectories (30 Epochs)
Normalized Pretraining Objective Loss
SSL Pretraining Epoch
SimCLR (NT-Xent)
DINO (Cross-Entropy Distillation)
MoCo (Queue InfoNCE)
BYOL (Cosine Distance)
(B) Two-Stage Downstream Fine-Tuning Convergence
Validation Accuracy (%)
Fine-Tuning Epoch
Stage 1:
Linear Head Warmup
(Backbone Frozen)
Stage 2:
Differential Fine-Tuning
(Cosine Annealing)
Top Model (DINO + EfficientNet-B0)
Scratch ResNet-50 (No SSL)